Sub-Queries

$q_1 \rightarrow$ SELECT * FROM TABLE_NAME ;

$q_2 \rightarrow$ SELECT " " table 2 ;

$q_1 (q_2)$

$q_1 \rightarrow$ Outer Query

$q_2 \rightarrow$ Sub-Query / INNER QUERY

- eg $\rightarrow q_2 \rightarrow$ results in one value $\rightarrow$ then q_1 executes based on that.

  $q_2 \rightarrow$ SELECT
  UPDATE
  DELETE

  [Not insert — as insert give any results]

$\rightarrow$ Sub-Queries mainly exist in 3 Queries:
- WHERE
- HAVING
- FROM

→ $q_1 (q_2 (q_3))$ → Possible ✓

Types of Subqueries

→ Clauses (place in a SQL Query)

→ Results of Inner Query

SELECT * FROM TABLE WHERE clause = ——

(select * from inner — ;)

SubQuery using SINGLE TABLE (with WHERE)

eg SELECT first_name, last_name FROM CUSTOMER

WHERE id IN (select id from customer WHERE gender="male");

NOT IN can also be used.

- SELECT avg (balance) from account;
- SELECT * from account (select avg (balance) from account);

WHERE balance >

using same table

Q: Find account balance of unblocked cards in the bank

```
SELECT * FROM card;
SELECT * from account;

SELECT balance from account;
SELECT * from card where is_blocked = 0;
```

Convert to sub-query

```
SELECT balance from account
where card_id = (select id from card
                 where is_blocked = 0);
```

Returns more than one error so error as equal to expects to return one row.

```
SELECT balance from account
where card_id != (select id from card
                  where is_blocked = 0);
```

opposite query

Derived Tables

Sub-queries → WHERE CLAUSE → Results of emergency

FROM CLAUSE — Singular Result set — < > <= (IN / NOT IN)

- **Derived Table** (AS)

Q: find the maximum value health insurance provided by the bank.

select * from Insurance where product_type = 'heath'; as temp_insurance

↓ eg Produces 2 rows

select max (product_value) from

ERROR: Every derived table must have its alias name

Co-Related Sub-Queries

Outer Query Operators (Sub Query)

↓

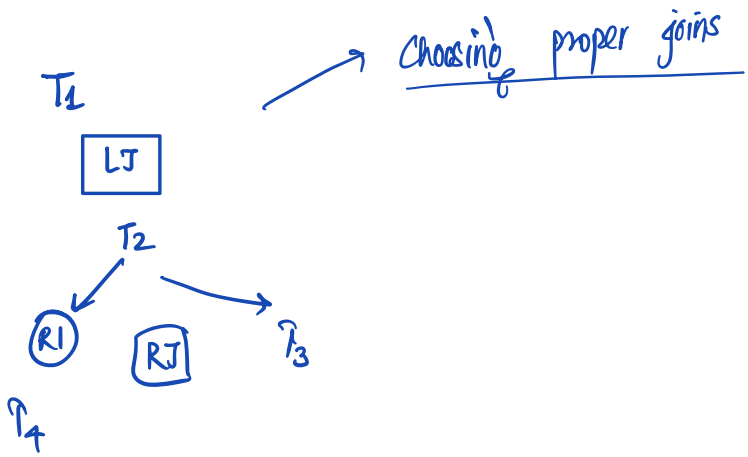
Inner Query — Runnable

Dependable

```
SELECT * FROM insurance;

SELECT
        product_name,
        product_value
FROM
        insurance n1

WHERE   product_value > (SELECT
                         AVG (product_value)
                         FROM INSURANCE
                         WHERE year_of_launch =
                         n1.year_of_launch);
```

Join vs Subqueries

- Structure of queries complex.
- Joins → ↓ Performance